

Question Paper

Exam Date & Time: 24-Dec-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU

SEVENTH SEMESTER B.TECH. DEGREE MAKEUP EXAMINATIONS - DECEMBER 2025

Natural Language Processing [NPT_4127]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Analyse the implications of skipping the 'Text Normalization' stage. How would this impact model training efficiency, model accuracy, and error patterns during testing? (4)
- A) Course Outcome: "CO1" / Cognitive Level: "Analysis"
- B) Using the rule shown (noun + -ly → adjective), apply the morphological transformation to the following nouns: love, cost, month. Write the resulting words and explain the meaning shift (4)
- Course Outcome: "CO1" / Cognitive Level: "Application"
- C) Apply syntactic analysis to the sentence 'The teacher discussed the problem with the student' by showing two possible parse structures and explain how each structure gives a different meaning. (2)
- (Assume the proper rules)
- Course Outcome: "CO1" / Cognitive Level: "Application"
- 2) Evaluate the effectiveness of using the Minimum Edit Distance algorithm to convert the misspelled word 'recieve' into the correct form 'receive' when the operation costs are insert = 3, delete = 3, and substitute = 5. (4)
- A) Course Outcome: "CO2" / Cognitive Level: "Evaluate"
- B) A system designer proposes the regular expression $^{(0[1-9][12][0-9][3[01]]/(0[1-9][1[0-2]]/[0-9][2])\$}$ to validate dates in the format dd/mm/yy. Evaluate the adequacy of this regular expression. Does it correctly capture all valid dates, and what limitations or edge cases reduce its reliability in real-world applications. (4)
- Course Outcome: "CO2" / Cognitive Level: "Evaluate"
- C) Explain how stems and affixes differ in their roles within word formation, and describe how each contributes to the overall meaning of a word with an example (2)
- Course Outcome: "CO2" / Cognitive Level: "Understand"
- 3) Evaluate the effectiveness of the Viterbi Algorithm by drawing the trellis diagram and predicting the most likely sequence of Part-of-Speech (POS) tags for the sentence: (4)
- A) "The dog chased the cat"
- Given the following parameters:
- States (Tags): Noun (N), Verb (V), Determiner (D)
- Transition Probabilities:

From \ To	N	V	D
N	0.10	0.70	0.20
V	0.40	0.10	0.50
D	0.85	0.10	0.05

Emission probabilities

Word	N	V	D
the	0.1	0.1	0.8
dog	0.85	0.10	0.05
chased	0.05	0.90	0.05
cat	0.85	0.10	0.05

Initial Probabilities:

$P(N) = 0.4$, $P(V) = 0.3$, $P(D) = 0.3$

Course Outcome: "CO3" / Cognitive Level: "Evaluate"

B) "The boy ran quickly to the park." (4)

Tag set: {DT, NN, VBD, RB, IN, JJ}

Hint rules:

1. Words like a, an, the → {DT}
2. Words ending with -ed → {VBD}
3. Nouns usually follow a determiner → {NN}
4. Words ending with -ly → {RB}
5. Common prepositions (at, in, on, with, to) → {IN}
6. Adjectives usually precede nouns → {JJ}

Apply rule-based tagging to tag each word.

Course Outcome: "CO3" / Cognitive Level: "Application"

C) Explain the bi-gram model with formulas. (2)

Course Outcome: "CO3" / Cognitive Level: "Application"

4) "The duck swims in the pond" vs. "I will duck under the table." (4)

A) Analyse how morphological, syntactic, and contextual knowledge help a POS tagger correctly tag the word "duck" in each sentence and which type of knowledge plays the most significant role in resolving such ambiguities.

Course Outcome: "CO4" / Cognitive Level: "Analysis"

B) For the given sentences: (4)

- Birds can fly in the sky
- Sky is blue during the day
- Day turns into night
- Night is dark and quiet

Evaluate a bi-gram model using this data. Show the probabilities of all the words and estimate the probability for the sentence "Birds can fly in the sky is blue during the day".

Course Outcome: "CO4" / Cognitive Level: "Evaluate"

C) Explain three methods of POS tagging. (2)

Course Outcome: "CO4" / Cognitive Level: "Understand"

5) A chatbot is designed using hand-crafted grammar. (4)

User: "I want to visit Paris."

A) **Bot:** "When do you want to visit Paris?"

User: "Next week."

Bot (fails): "Next week what?"

Evaluate why hand-crafted grammar rules fail to interpret elliptical replies like "Next week," and argue whether a rule-based system could ever fully handle such dialogue ellipsis in real-world contexts.

Course Outcome: "CO5" / Cognitive Level: "Evaluate"

B) Consider the following CNF grammar: (4)

S → **NP VP**

NP → **Det N**

NP → **NP PP**

VP → **V NP**

VP → **VP PP**

PP → **P NP**

Det → "the" | "a"

N → "student" | "professor" | "binoculars"

V → "watched"

P → "with" | "near"

Examine the CKY parse table for the sentence '*The professor watched the student with binoculars*' and identify the cell where PP-attachment ambiguity arises.

Course Outcome: "CO5" / Cognitive Level: "Analysis"

C) Explain the difference between dependency grammar and constituency grammar and how each framework models sentence structure differently. (2)

Course Outcome: "CO5" / Cognitive Level: "Understand"

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Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026 NPT_4127:Natural Language Processing

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

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058-MIT BLR-MAKEUP EXAM-2025-2026 NPT_4127:Natural Language Processing

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	3	6(12%)
3	Application	4	12(24%)
4	Analysis	3	12(24%)
5	Create	5	20(40%)

6	Evaluate	0	0(0%)
	Total	15	50 (100%)